The Knowledge Complexity of Quantum Problems

Giulio Malavolta Bocconi University

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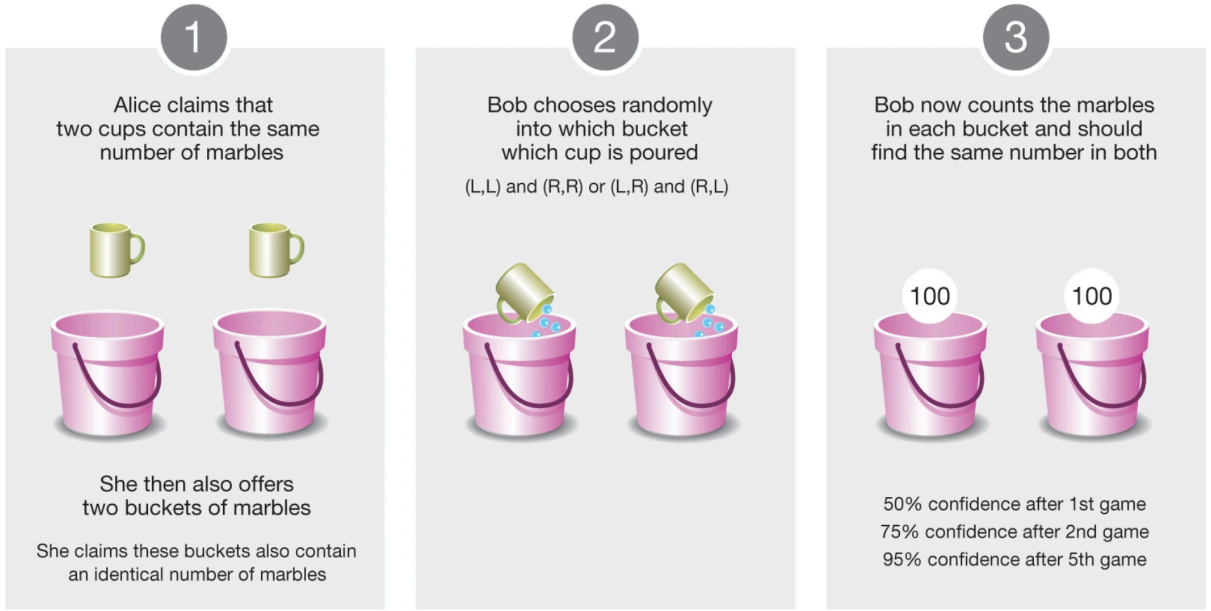
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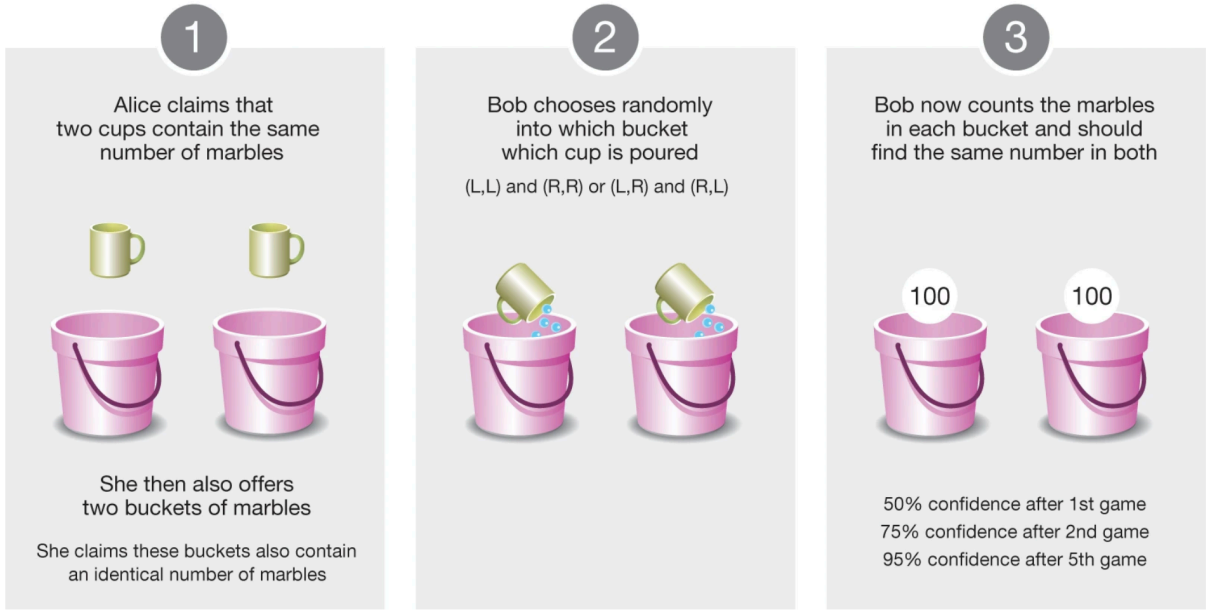


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Can we prove **any physical fact** in zero-knowledge?

Quantum promise problems

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A **quantum promise problem** is a pair $X = \left(X_{\text{yes}}, X_{\text{no}}\right)$ of disjoint sets of quantum states.

The instance is a state ρ, handed out as copies $\rho^{\otimes \text{poly}(\lambda)}$.
Pure states define the classes pQMA, pQIP, …; mixed states define mQMA, mQIP, …

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Same marginal?
Given $\left|\psi\right\rangle_{AB}$ and $\left|\phi\right\rangle_{AB}$: is
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Unlike, say, a QMA statement, these are claims about *a particular* state, not about the existence of a witness.

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$m = 1$: pQMA, mQMA (the message is the witness) · 2/3 and 1/3 are arbitrary [CCHS24]

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1 **Zero-knowledge for every quantum problem in pQIP**

If EFI pairs exist, then $\mathsf{pQIP} \subseteq \mathsf{pQIP}_{\mathrm{zk}}$.

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2 **No zero-knowledge for mixed states**

If the learning with errors (LWE) problem is hard, there exist problems in mQMA without an efficient zero-knowledge protocol.

Motivation

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Zero-knowledge quantum property testing

Property testing asks how many copies of ρ decide whether it has a property [BFNR08, MdW13].

Can an untrusted prover lower the copy complexity [CCHS24]? With zero-knowledge, even a malicious verifier learns nothing beyond the answer.

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Zero-knowledge Uhlmann

[BEM+23] identified the **Uhlmann transformation problem** as a central primitive for:

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2. quantum cryptography
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Quantum cryptography

A [GMW91]-like compiler for quantum protocols?

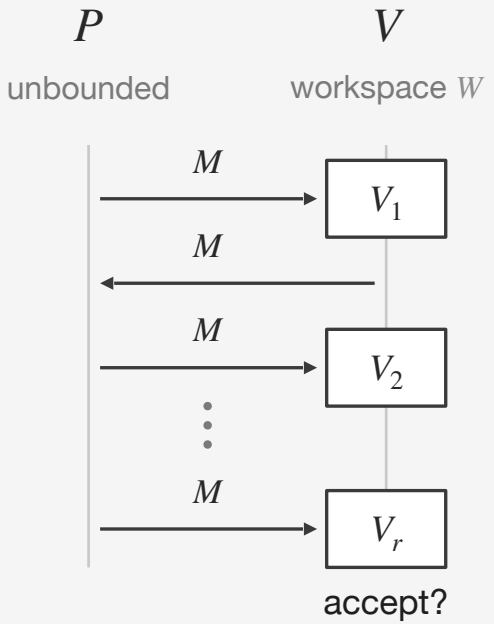
Example: the verifier holds $\mathrm{Com}(C)$ and copies of $|\psi\rangle$; prove that $C|0\rangle = |\psi\rangle$.

Zero-knowledge for quantum problems gives a systematic approach.

Honest-verifier zero-knowledge

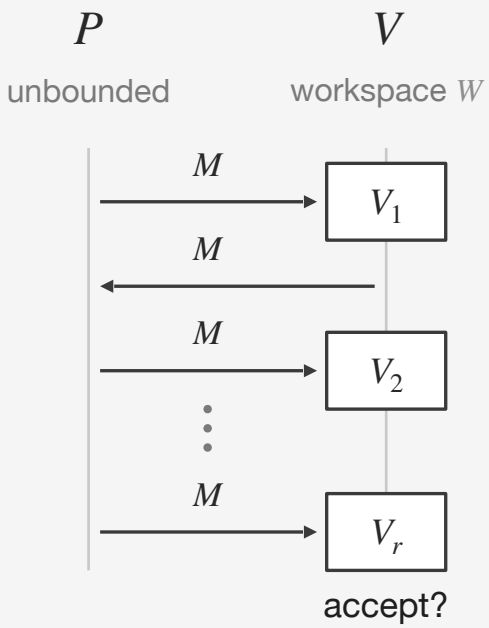
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Interactive proof

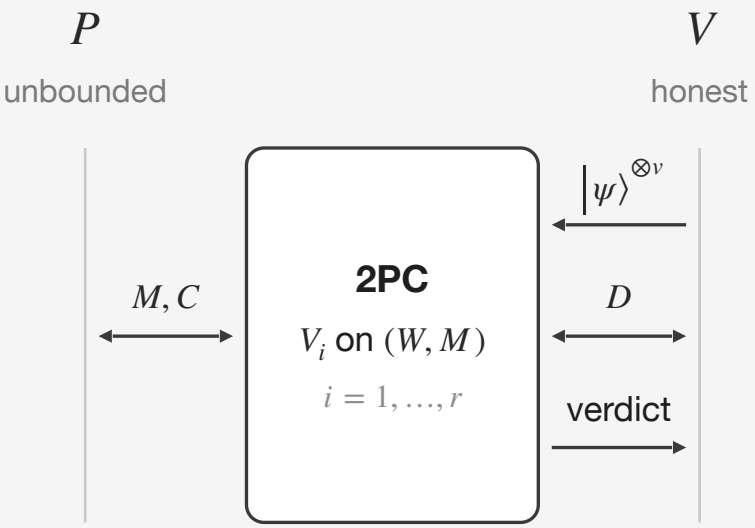


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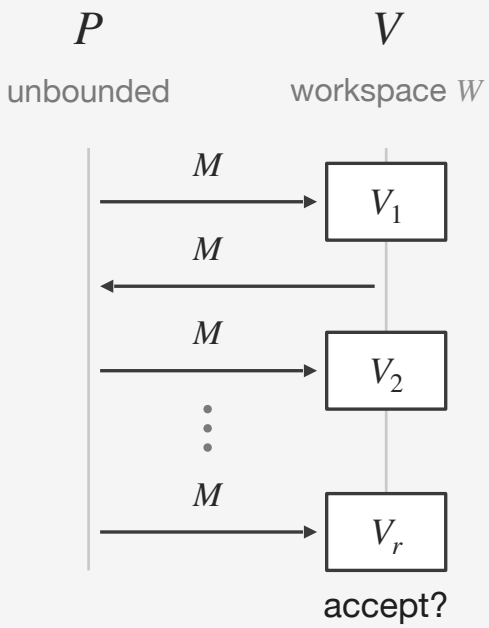


Honest verifier

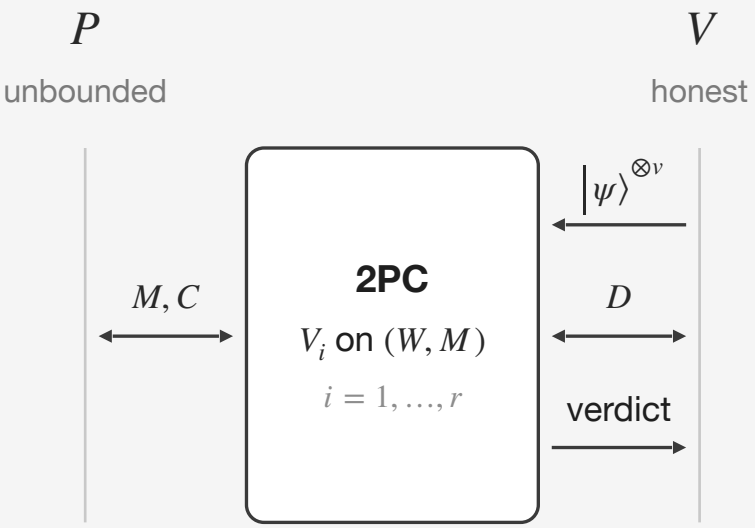


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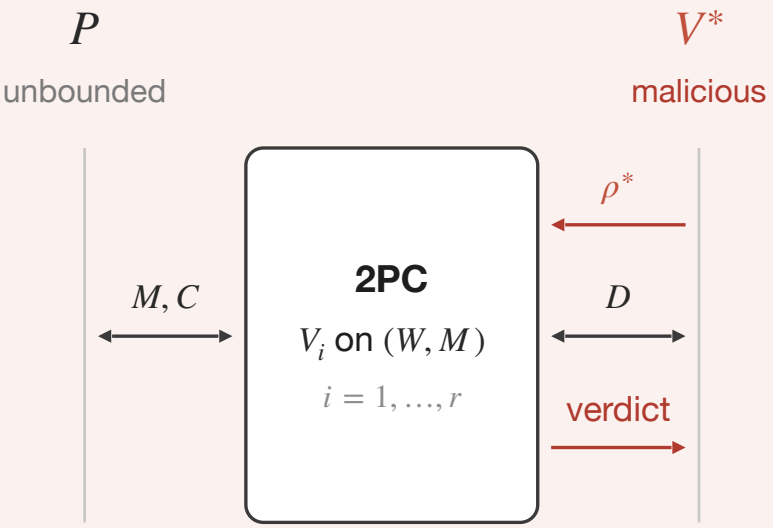
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Honest verifier



Malicious verifier

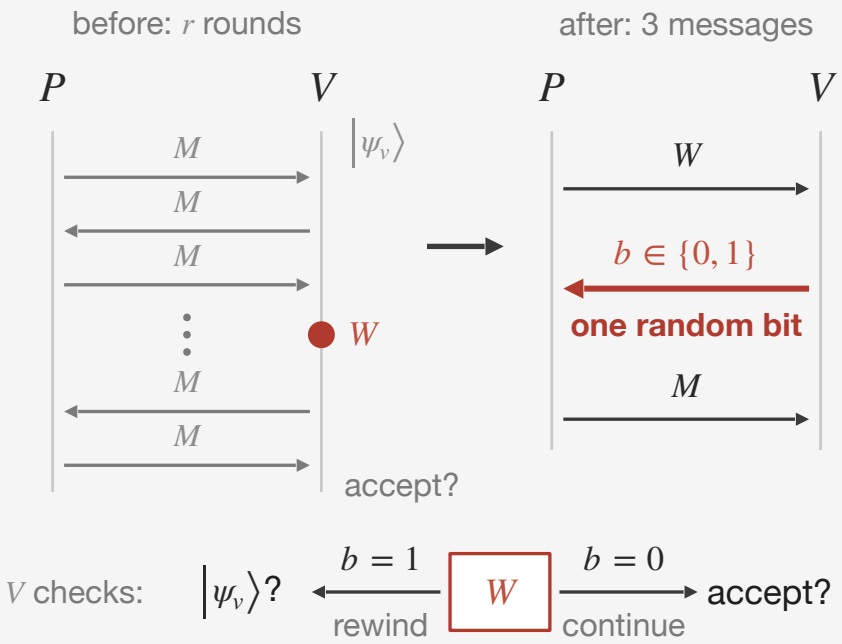


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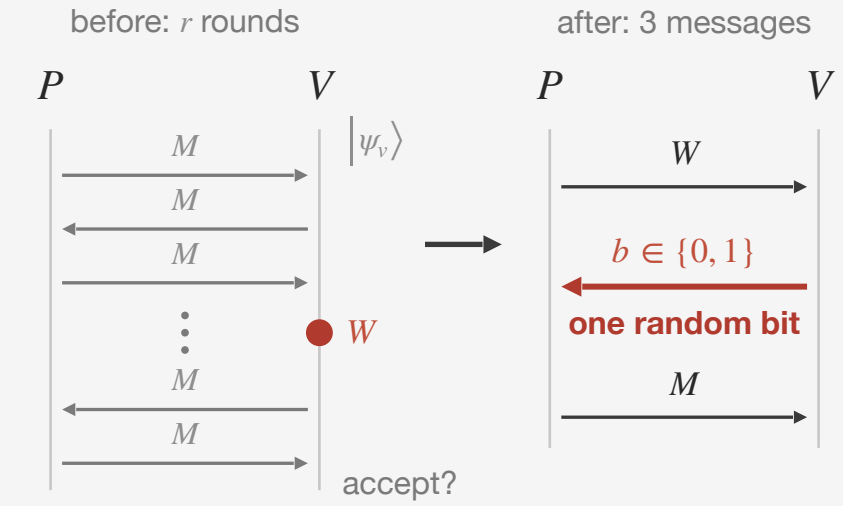
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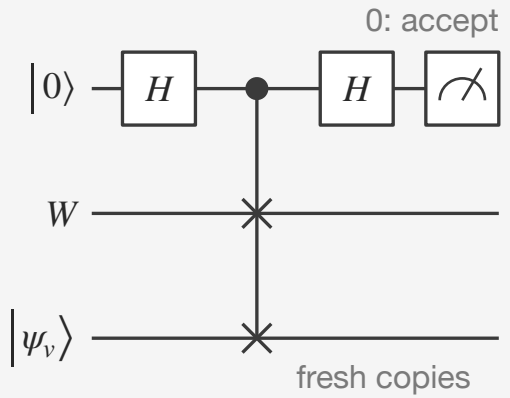
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2 SWAP test

V cannot project onto $|\psi_v\rangle$

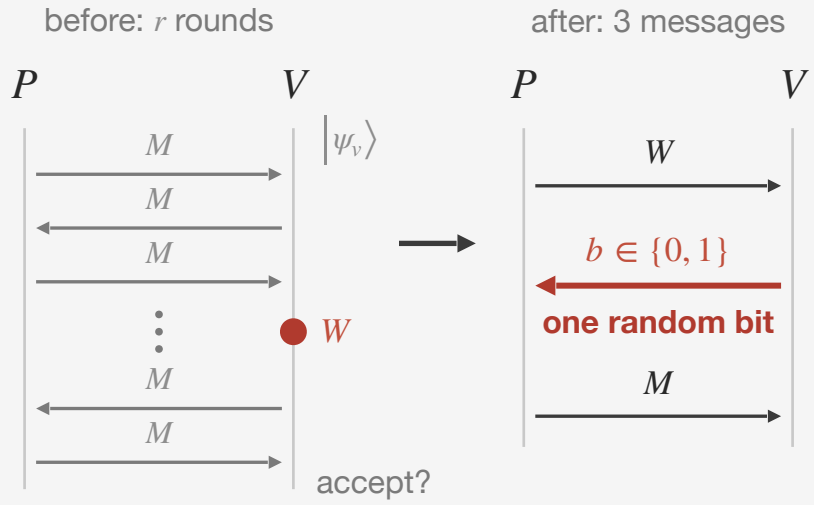


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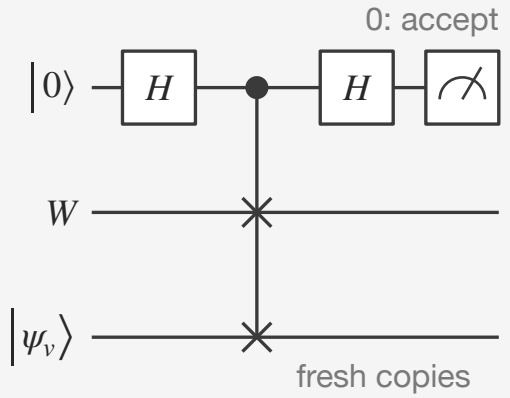
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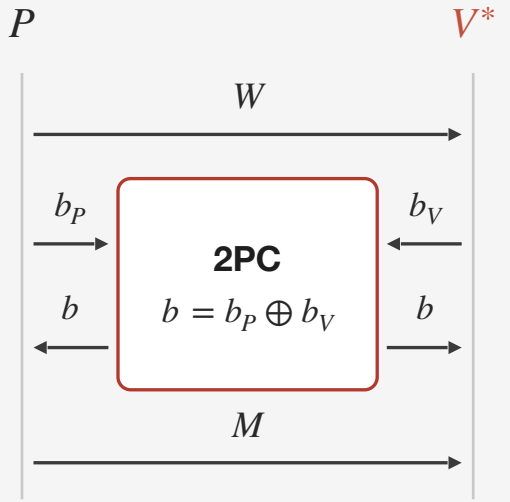
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3 Coin toss in a 2PC

V only sends the coin b



V^* cannot bias b; the simulator programs it: no rewinding

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Questions?

https://arxiv.org/abs/2510.06923